FINAL PROJECT • AI IN CONSTRUCTION AND FACILITIES

AI Adoption in Construction and Facilities

A qualitative study of AEC industry practices

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Essential Question: How is Artificial Intelligence being adopted, implemented, and managed within the architecture, engineering, and construction (AEC) industry?

Purpose: connect theory with practice

The project compared interview findings with academic frameworks on AI adoption.

Academic lens

TAM	Usefulness, ease of use, and workflow fit
Maturity	Readiness, training, staged implementation
Governance	Security, privacy, risk, and human oversight

Core idea: AI adoption is not just buying software. It is organizational change.

Study goals

- Identify AI tools and use cases
- Compare adoption strategies across organizations
- Analyze drivers, barriers, and readiness
- Develop practical recommendations

Four perspectives across three organizations

Participants are reported anonymously because explicit naming consent was not confirmed for all interviews.

AEC + Facilities AI Adoption

Healthcare Organization

IT Director

Owner / facilities-side view

General Contractor A

Construction Technology Director

Tech strategy + field tools

General Contractor A

Project Manager

PM workflows + Procore/Copilot

General Contractor B

Senior VP / Technology Leader

Leadership + process improvement

AI is already embedded in everyday workflows

Most current value appears in document-heavy, meeting-heavy, and information-search tasks.

Tool	Use cases
Microsoft Copilot	meeting minutes • email chains • document review
Procore AI / Read AI	spec search • action items • PM documentation
Epic + Ambient Documentation	provider messaging • clinical documentation
DroneDeploy / Reality Capture	360 walks • safety observations • progress tracking
Custom AI Agents	help desk tasks • on-call lookup • process Q&A

Pattern: tools were selected because they fit existing platforms and workflows, not because AI was trendy.

AI adoption is problem-driven

Participants rejected “flashy” technology unless it solved a real workflow problem.

AI was viewed as a tool to reduce pain points — not as a solution by itself.

1. **Find the pain point**
   subcontract review, specs, tickets, minutes
2. **Pilot a focused tool**
   Copilot, Procore AI, agents, safety AI
3. **Measure workflow value**
   time saved, faster answers, fewer tickets
4. **Scale carefully**
   training, governance, human review

Governance, security, and trust shape adoption

Risk controls were not side issues — they were central selection criteria.

Healthcare organization

AI governance committee • HIPAA / PHI guardrails • pilots before scaling

General contractors

proprietary data protection • approved platforms • human validation of output

Shared pattern

leadership involvement • low-risk first use cases • confidence before rollout

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estimated hours/week saved by two internal healthcare IT agents

Measured value helped build the case for continued AI adoption.

Office workflows are benefiting first

Field-side AI is promising, but still harder to implement consistently.

CLEARER CURRENT VALUE

Office / process workflows

- Meeting documentation
- Email and document summaries
- Spec and drawing search
- Subcontract review
- Internal process Q&A

EMERGING NEXT FRONTIER

Field intelligence

- 360 walks and drones
- PPE / safety observations
- BIM-to-field progress tracking
- Subcontractor coordination
- Cleaner data needed

Why the gap? Field work is physical, conditions change quickly, data can be messy, and the consequence of errors is higher.

Barriers were consistent across interviews

The strongest success factors were not technical features — they were implementation choices.

Primary challenges

- Data security and privacy
- Trust in AI outputs
- Training time and cost
- Field adoption resistance
- Messy or incomplete data

Success factors

- Leadership support
- Pilot teams and champions
- Workflow integration
- Human oversight
- Clear KPIs and feedback loops

A practical adoption playbook

Recommended steps for construction and facilities organizations beginning or expanding AI use.

1
Build governance first
include leadership, IT, users, risk/compliance

2
Pick high-value use cases
start with meetings, specs, documents, tickets, safety observations

3
Pilot with champions
select tech-forward teams and collect feedback

4
Measure value
track time saved, ticket reductions, quality, safety observations

5
Scale with human oversight
train users and require review for high-risk outputs

Implementation message: start small, prove value, then scale responsibly.

AI will augment construction expertise — not replace it.

Across interviews, human review remained essential for clinical, safety, contractual, and field-related outputs.

Agents — more internal process automation and document review

Reality capture — AI safety observations and progress tracking

Governance — data rules will become a competitive advantage

Bottom line: organizations with strong governance, clean workflows, and trusted users will move fastest.

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Interview data collected by the author from anonymized construction and healthcare professionals, April-May 2026.

Note: Interview participants are anonymized due to uncertain explicit consent for naming employees and organizations.